Arista Access Node KR-ACC-56 Optical Transport Failure SOP (ALARM_5192)

Node: KR-ACC-56 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_5192

Description

This Standard Operating Procedure (SOP) outlines the diagnostic and resolution steps for a major optical transport failure on Arista Access Node KR-ACC-56 at KR Puram Access. The failure domain encompasses SFP/XFP degradation, fiber cuts, and Rx/Tx power loss. The underlying protocol mechanics include the utilization of Small Form-factor Pluggable (SFP) and Extended Form-factor Pluggable (XFP) modules for optical signal transmission and reception, which are susceptible to degradation due to environmental factors or manufacturing defects. Furthermore, optical signal attenuation, caused by fiber cuts or bending losses, can lead to reduced signal quality, resulting in Rx/Tx power loss and subsequent alarm triggering. This SOP aims to identify the root cause of the failure and implement corrective actions to restore the network to a stable state.

Pre-requisite Checks

- Verify that the KR-ACC-56 node is reachable via console or SSH.

- Confirm that the fiber optic cables are securely connected to the SFP/XFP modules and that there are no signs of physical damage.

- Check the redundant link status to ensure that it is operational and can take over in case of a primary link failure.

- Dispatch remote hands to KR Puram Access to visually inspect the fiber optic cables and SFP/XFP modules.

Potential Root Causes

Cause	Probability
SFP/XFP degradation or failure	High
Fiber optic cable damage or cut	Medium
Rx/Tx power loss due to optical signal attenuation	Low

Diagnostic Commands & Expected Outputs

```
Command: show interface e1/1 detail, Expected Output: Port 1 is up, line protocol is up (connected), Description: This command is used to verify the status of the port and confirm whether it is up or down. The expected output indicates that the port is up and the line protocol is up, which suggests that the optical signal is being transmitted and received correctly.
```

```
Command: show sfp e1/1, Expected Output: SFP-10G-SR (Serial Number: ABCDEFGHIJK), Description: This command is used to retrieve information about the SFP module connected to the port. The expected output indicates that the SFP module is a 10G-SR type with a specific serial number.

Command: show sfp e1/1 detail, Expected Output: Optical signal quality: OK, Description: This command is used to retrieve detailed information about the SFP module, including the optical signal quality. The expected output indicates that the optical signal quality is OK, suggesting that there are no issues with the SFP module.
```

Resolution Steps

Step 1: Step: Replace the SFP/XFP module, Command: replace sfp e1/1, Description: This command is used to replace the SFP/XFP module with a new one. The new module should be of the same type and speed as the original module.

Step 2: Step: Verify the fiber optic cables, Command: show interface e1/1 detail, Description: This command is used to verify the status of the port and confirm whether it is up or down. If the port is still down, it may indicate that there is an issue with the fiber optic cables.

Step 3: Step: Replace the fiber optic cables, Command: replace interface e1/1, Description: This command is used to replace the fiber optic cables. The new cables should be of the same type and length as the original cables.

Post-Check Verification

```
Command: show interface e1/1 detail, Expected Output: Port 1 is up, line protocol is up (connected), Description: This command is used to verify the status of the port and confirm whether it is up or down. The expected output indicates that the port is up and the line protocol is up, which suggests that the optical signal is being transmitted and received correctlv.

Command: show sfp e1/1 detail, Expected Output: Optical signal quality: OK, Description: This command is used to retrieve detailed information about the SFP module, including the optical signal quality. The expected output indicates that the optical signal quality is OK, suggesting that there are no issues with the SFP module.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: replace sfp e1/1, Description: This command is used to restore the original SFP/XFP module.

- Step: Restore the original fiber optic cables, Command: replace interface e1/1, Description: This command is used to restore the original fiber optic cables.